H109 — torsion-local division-polynomial derivative oracle

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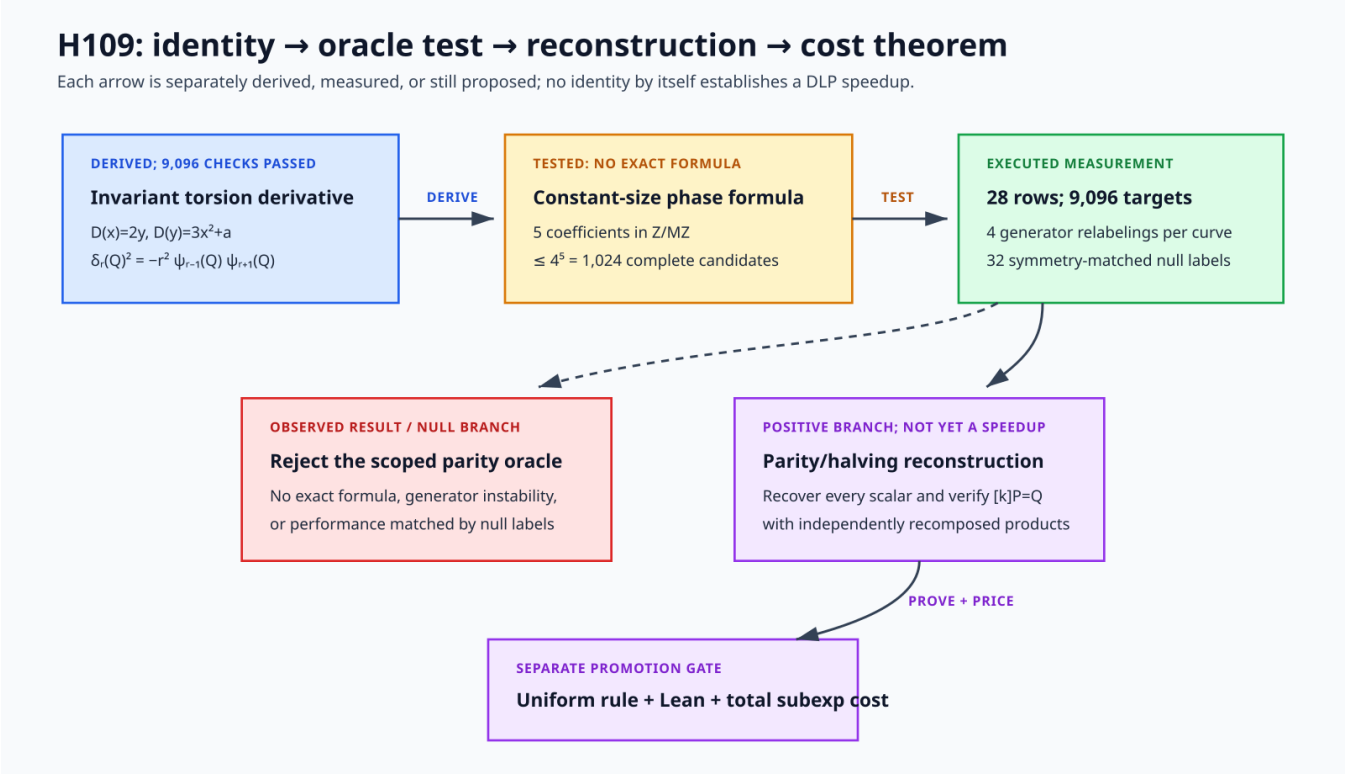


Figure 1: H109 evidence flow

Editable vector source: `h109-torsion-derivative-flow.svg`.

The diagram separates four claims: the local algebraic identity, existence of an exact compact phase formula, end-to-end scalar recovery, and a uniform subexponential cost theorem. Only the first is mathematically predicted at freeze time; all other arrows are proposed or pending measurement.

Requirement-to-evidence contract

- **Materially new mechanism:** use the invariant local derivative `D psi_r(Q)`, not a fitted span of character codewords.
- **Novelty screen:** preserve database queries, closest sources, and a scoped "novelty not established" conclusion.
- **Measurement:** evaluate every target on seven curves and four generator relabelings; report field, recurrence, character, and decoder costs.
- **Control:** use the paired H107/H108 baseline, local-identity checks, and symmetry-matched random parity labels with the same compact formula budget.
- **Proof:** provide Lean certificates for the derivation, phase/parity interface, and binary reconstruction. A complexity theorem is required for any subexponential claim.
- **PR:** open one branch and one draft PR for H109 before implementation.

Mechanism

For the invariant derivation on a short Weierstrass curve

```
D(x) = 2y,
D(y) = 3x^2 + a,
```

let

delta_r(Q) = D(psi_r)(Q), Q in <P> minus {O}.

Every nonzero subgroup point is a simple zero of psi_r. Comparing the Laurent expansion of the multiplication formula

x([r]R) = x(R) - psi_(r-1)(R) psi_(r+1)(R) / psi_r(R)^2

with [r]^*(dx/(2y)) = r dx/(2y) predicts the local identity

delta_r(Q)^2 = -r^2 psi_(r-1)(Q) psi_(r+1)(Q).

H109 first checks this identity exactly at every frozen point. It then tests a constant-size family of multiplicative character formulas made from

delta_r(Q), psi_(r-1)(Q), psi_(r+1)(Q), y(Q), constant primitive root.

Unlike H105–H108, the number of fitted coefficients is bounded independently of r. Division-polynomial values and invariant derivatives are evaluated by memoized recurrence with dual-number/product-rule arithmetic.

Frozen formula family and protocol

Let M=2^v2(p-1) and encode maximal base-field character phases in Z/MZ. Completely enumerate every coefficient tuple

(c0,c_delta,c_minus,c_plus,c_y) in (Z/MZ)^5

for the phase of

```
g^c0 delta_r^c_delta psi_(r-1)^c_minus
    psi_(r+1)^c_plus y^c_y.
```

This is at most 4^5=1024 formulas on the frozen curves. No coefficient budget grows with r, and no heuristic search is permitted.

1. Run all seven retained curves, generator multipliers 1,2,3,5, and every nonzero subgroup point.
2. Check psi_r(Q)=0, nonvanishing of the four formula inputs, the predicted squared-derivative identity, and agreement between memoized dual recurrence and a direct small-index symbolic/difference control.
3. Score every frozen formula against canonical scalar parity. Retain exact formulas, best matches, all ties, coefficient vectors, and operation counts.
4. Run 16 symmetry-matched random parity-label controls on the primary p=1 mod 4 row E101 and 16 on the primary p=3 mod 4 row E127. Each control assigns complementary bits to k and r-k and uses the identical complete formula family.
5. Compare with H107/H108's exact but linear-support character modules, charging their setup and support rather than only online character calls.
6. If any exact observed formula exists, execute the parity/halving decoder on every target for that row. A primary positive must replicate across all four generators; the generic gate requires all 28 rows.
7. Independently reevaluate any exact formula through a separately composed field product and validate every reconstructed scalar by EC multiplication.

Runtime and proof gates

The candidate qualifies as subexponential only if all of the following are established:

- a public coefficient rule or the frozen constant-size enumeration constructs an exact formula on every generic row;
- memoized evaluation of the required values and derivatives has a proved poly(log r, log p) or L_r(1/2,O(1)) bound including field operations;
- the parity decoder is exact and its total setup, advice, online work, memory, and verification stay within that bound;
- Lean proves the certificate-to-parity step, reconstruction, and the cost theorem actually used by the claim.

An exact toy formula without the uniform theorem is evidence on the frozen instances, not a generic subexponential algorithm.

Prior-art screen

Queries were run on Crossref, OpenAlex, and arXiv for combinations of "division polynomial derivative," "torsion point," "elliptic divisibility sequence derivative," "theta group local parameter," and "discrete logarithm." Closest located sources were:

- Coombes, "Elliptic curves and logarithmic derivatives";
- Silverman, "p-adic properties of division polynomials and elliptic divisibility sequences";
- Burhanuddin and Huang, "Elliptic curve torsion points and division polynomials";
- Lauter and Stange's EDS/ECDLP equivalence.

The bounded search did not locate the frozen derivative-phase parity formula, but it is not source-complete. Status: mechanism-distinct; novelty not established.

The repository lead-atlas screen also found the closest internal predecessor, ECDLP-IDEA-006, and its independent P1540 audit. Those records test low-rank/annihilator and translated-coordinate index location; the audit shows that the QRT state is birationally conjugate to the original elliptic translation and that low-degree rational additive or multiplicative linearization is unavailable. H109 does not claim an annihilator or a new state representation: its separately falsifiable question is whether the *torsion-local invariant derivative* has a constant-size two-power character phase that outputs scalar parity. A positive formula must therefore survive the complete formula enumeration, generator relabelings, matched random-label controls, full scalar reconstruction, and total-cost gate below. This distinction is mechanism-level only and does not establish novelty.

PR gate

Implementation may start only after this freeze is present in a draft H109 PR. The canonical experiment PR belongs in `isogeny-labs/crypto`; the paired reference-harness PR belongs in `isogeny-labs/cryptanalysis`. Neither PR is a result until the frozen matrix has run and its artifacts are committed.

Executed experiment

The draft gates were opened as `isogeny-labs/crypto#3102` and `isogeny-labs/cryptanalysis#1342`. Implementation commit `eb6cc02` then executed the protocol without changing the coefficient family, curve cohort, target set, controls, or stopping rule.

All 9,096 target rows passed every algebraic control:

- `psi_r(Q)=0` and all four candidate inputs were nonzero;
- all squared local identities had residue zero;
- the memoized values agreed with the independent sequential recurrence;
- derivatives `D psi_2` through `D psi_8` agreed with independently differentiated symbolic polynomials at every target.

The evaluator materialized 412,328 non-base recurrence nodes (26–54 per target as `r` grew from 83 to 947), used 10,674,040 counted field multiplications and 211,712 inversions, and retained at most a small logarithmic-size cache per target. This validates the fast evaluator on the frozen cohort; it does not supply a DLP oracle.

The complete formula searches examined 8,832 coefficient tuples across the 28 rows. **No exact formula occurred on any row.** Best agreement ranged from 0.500000 to 0.611111, with median 0.530826. The 32 symmetry-matched controls ranged from 0.500000 to 0.629630, with median 0.509259, and likewise had no exact formula. On the two frozen primary comparisons:

```
E101: candidate 0.500000; control mean 0.500000; empirical tail p=1.000000.
E127: candidate 0.574074; control mean 0.560185; empirical tail p=0.411765.
```

The empirical tail values use the conservative plus-one rank among 16 frozen controls. They are descriptive finite-panel statistics, not asymptotic probabilities. Because no row was exact, the preregistered parity/halving decoder gate correctly did not open.

Diagnostic wall totals in the committed artifact were 6.3530 s for candidate evaluation, 0.7493 s for formula enumeration, and 15.1463 s for the deliberately expensive independent symbolic/value controls; the complete Python process took 25.7005 s in the recorded container. These are unisolated diagnostic times, not performance benchmarks. The generic square-root references and H107/H108 artifact hashes remain in the raw result for accounting.

A second complete run changed only diagnostic timing fields. After removing clock and environment fields, both runs had scientific-content SHA-256 `3991b25d2a2ddac0ebb1d25370c8b110051a40acca64a6bda3dd3baba4191b69`.

Lean 4.19.0 checked the tangent-derivation lemma, multiplicative-character transport, exact phase-to-parity interface, binary reconstruction, and conditional cost composition. `#print axioms` reports only the named standard Lean foundations for the applicable theorems. The actual division-polynomial local identity remains a written derivation plus exhaustive frozen check, not a fully formalized division-polynomial theorem.

Result: the local identity and polylogarithmic-node evaluator are supported on the frozen instances, but the constant-size phase-to-parity bridge is rejected for the complete preregistered family. This closes H109 only; it neither rules out other functions of torsion-local data nor establishes a generic ECDLP lower bound. The next experiment must use a mechanism other than widening this five-input multiplicative character formula.

Committed result SHA-256: `940123c44c6194b247fab3596a9ed2e331ee7e01b2fb8830fc7e8465e354993a`.